Algorithm for file updates in Python

Project description

As a security professional , I need to regularly update a file that identifies the employees who can access restricted content in my organization.There is an allow list for IP addresses permitted to sign into the restricted subnetwork. There's also a remove list that identifies which employees you must remove from this allow list. . To perform this task, I have created the algorithm that automates the task by updating the file “allow_list.txt” after removing the IP addresses that should no longer have access.

Open the file that contains the allow list

```
# Assign `import_file` to the name of the file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Display `import_file`

print(import_file)

# Display `remove_list`

print(remove_list)
```

The allow_list.txt contains the IP addresses allowed to access the contents and it is assigned on the variable “import_file”. the remove_list contains the IP addresses that should not access the contents. In this code, both the import file and remove_list are printed on the screen using print() function.

Read the file contents

```
# Assign `import_file` to the name of the file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file

with open(import_file, "r") as file:

  # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`

  ip_addresses=file.read()

# Display `ip_addresses`

print(ip_addresses)
```

Now , we need to read the file contents inside the import_file using the with statement. The with handles the error and manages the external resources and open() is used to open a file with two parameters including the file to open i.e.,import_file and "r" to read the file. Outside the with statement, the "as file" is used to save the contents within the with statement in file. Inside the body of the with statement, the contents of the file is assigned on the ip_addresses variable. Afterwards, the ip_addresses are displayed using print().

Convert the string into a list

```
# Assign `import_file` to the name of the file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file

with open(import_file, "r") as file:

  # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`

  ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()

# Display `ip_addresses`

print(ip_addresses)
```

To convert the string into a list, we use a .split() function to split the contents inside the ip_addresses to a list based on the whitespaces.now the ip_addresses is displayed as the list.

Iterate through the remove list

```
# Assign `import_file` to the name of the file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file

with open(import_file, "r") as file:

  # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`

  ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()

# Build iterative statement
for elements in ip_addresses:



    # Display `element` in every iteration

    print(elements)
```

Now , we have to iterate through the remove_list which consists of IP addresses that are no longer to access. To iterate it, we use a for loop using a loop variable "element" iterating through remove_list and the elements are displayed on the screen.

Remove IP addresses that are on the remove list

```
import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file

with open(import_file, "r") as file:

  # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`

  ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable `element`
# Loop through `ip_addresses`

for element in ip_addresses:

  # Build conditional statement
  # If current element is in `remove_list`,
    if element in  remove_list:
        # then current element should be removed from `ip_addresses`
        element=ip_addresses.remove(element)

# Display `ip_addresses`

print(ip_addresses)
```

Now , let's remove the IP addresses that are no longer accessible to the restricted contents. In the previous code, we use a for loop to iterate through the ip_addresses. Inside the for loop, an if statement is used with a condition that the loop variable element is in the remove_list. If it is, then remove the element from the ip_addresses and assign it to the element. Afterwards, display the ip_addresses.

Update the file with the revised list of IP addresses

```
# Assign `import_file` to the name of the file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file

with open(import_file, "r") as file:

  # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`

  ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable `element`
# Loop through `ip_addresses`

for element in ip_addresses:

  # Build conditional statement
  # If current element is in `remove_list`,

    if element in remove_list:

        # then current element should be removed from `ip_addresses`

        ip_addresses.remove(element)
```

```
# Convert `ip_addresses` back to a string so that it can be written into the text file

ip_addresses = " ".join(ip_addresses)

# Build `with` statement to rewrite the original file

with open(import_file,"w") as file:
    file.write(ip_addresses)
```

Now we need to update the file with the revised list after removing the IP addresses that were no longer accessible to the contents. In order to perform these, we need to first convert the ip_addresses to string using the .join() method. The updated file is written to the import_file using the with statement using the "w" parameter and .write() method inside the with statement.

Summary

Checked the allowed_list and removed_list files and opened the allowed_list.txt file using the with statement and converted into list using the .split() method which is assigned to ip_addresses list. Using a for loop , we iterate through each element in the list and check whether the IP addresses in the remove_list are present in the ip_addresses list and remove it from the list. Update the file revised list of IP addresses.